Rethinking Noisy Correspondence via Cross-Modal Collaborative Representation

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Abstract

Cross-modal retrieval aims to establish semantic alignments across different modalities. However, existing methods frequently assume the availability of perfectly paired training data. This assumption ignores the inevitable noisy correspondence (NC) within real-world datasets, which degrades the performance of retrieval. To address these limitations, we propose the Multi-phase Collaborative Representation (MCR) framework. To the best of our knowledge, MCR is the first framework that introduces cross-modal collaborative representation to tackle the NC problem, accurately identifying true correspondences. Specifically, MCR reconstructs a query feature using a linear combination of training samples within the batch from the other modality. By optimizing the objective of collaborative representation, we leverage the reconstruction coefficients to measure the semantic correlation between each pair. Furthermore, we propose a multi-phase iterative process that enhances the feature representations of clean samples and dynamically filters noisy ones. This process produces reliable matching confidence weights to effectively quantify the noise. Extensive experiments on three widely-used benchmark datasets, including Flickr30K, MS-COCO, and Conceptual Captions, validate the effectiveness of our proposed approach. The code is available at https://github.com/Ji-Xiaoying/MM26_MCR.

CCS Concepts

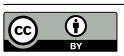
• **Information systems** → **Multimedia and multimodal retrieval**.

Keywords

Cross-Modal Retrieval, Noisy Correspondence Learning, Collaborative Representation

ACM Reference Format:

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1 Introduction

Cross-modal retrieval (CMR) [4, 5, 20, 24, 38] aims to establish correspondence and alignment between data from different modalities, such as images and text, or images and audio. Although this field has made significant progress, most existing methods assume that the training data is correctly aligned across modalities. In real-world scenarios, constructing high-quality multi-modal datasets with human-annotated alignments, such as Flickr30K [44] and MS-COCO [23], is extremely labor-intensive and time-consuming. To reduce annotation costs, many studies collect data directly from the Internet. However, datasets obtained through crowd-sourcing or web crawling, such as Conceptual Captions [35], inevitably treat many semantically mismatched data pairs as matched pairs due to the lack of strict cleaning and filtering mechanisms. This issue introduces the noisy correspondence (NC) problem. Training CMR models directly on such noisy datasets disrupts the alignment of modal representations in the feature space, leading to a degradation in the performance of cross-modal retrieval [8, 14, 18, 42].

To address the noisy correspondence problem, most mainstream approaches adopt established paradigms from the domain of noisy labels [37, 41, 45]. These approaches [10, 17, 18, 46] utilize the memorization effect of Deep Neural Networks (DNNs) [9], where models tend to prioritize fitting simple and clean samples during the early stages of training. Guided by this phenomenon, researchers usually leverage binary mixture models, such as the Gaussian Mixture Model (GMM) [7, 29] or the Beta Mixture Model (BMM) [42, 43], to model the similarity or loss distribution of sample pairs. Then these methods partition the training data into clean and noisy subsets via fitted distributions for subsequent correction. For instance, Bicro [42] estimates the reliability of samples by utilizing the inherent similarity between modalities. ESC [43] quantifies noisy content by analyzing the rate of change in similarity. NPC [50] assigns sample noise levels by evaluating the negative impact of each sample.

Nevertheless, existing studies essentially inherit and build upon denoising strategies originally designed for noisy labels. Such a direct paradigm transfer overlooks the intrinsic characteristics of cross-modal data. Rather than exploiting the rich collaborative relationships within the cross-modal feature space, these methods reduce multi-modal data into simple distributions of similarity or

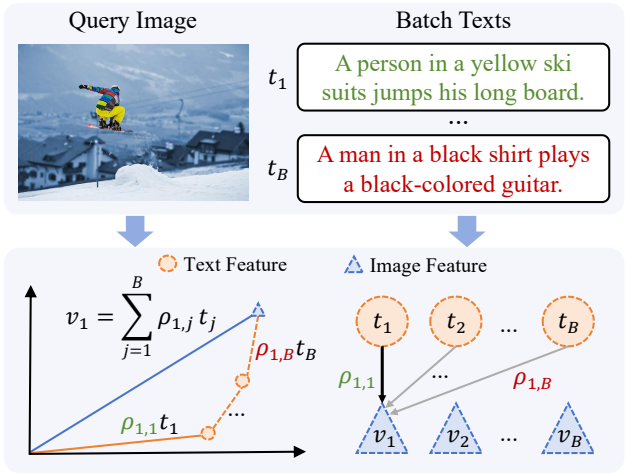


Figure 1: Illustration of MCR. A query image feature is linearly reconstructed using a dictionary formed by batch text features. The resulting reconstruction coefficients reflect the semantic contribution of each text, which are utilized to quantify the matching confidence of the image-text pairs.

loss to distinguish clean pairs from noisy ones. Consequently, the inefficient exploitation of feature information causes the distributions of clean and noisy samples to inevitably overlap, making them difficult to accurately separate. Especially under high-noise environment, this limitation restricts the ability of the model to identify accurate correspondences.

To overcome these limitations, we introduce the theory of collaborative representation [48, 49] into the cross-modal domain to re-establish the feature alignment process. As illustrated in Fig. 1, the core idea is to treat the data from one modality as a rich semantic dictionary. Within an ideal cross-modal feature space, a query sample from the other modality can be collaboratively and linearly reconstructed using this dictionary. The reconstruction coefficients adaptively quantify the actual semantic correlation between samples, therefore it can be a reliable signal for matching confidence. Motivated by this insight, we design a solution via cross-modal collaborative representation to tackle the noisy correspondence problem, thereby achieving robust cross-modal retrieval.

To this end, we propose an efficient approach called **M**ulti-phase **C**ollaborative **R**epresentation (**MCR**). This approach constructs a multi-phase sample evaluation and filtering framework that deeply mines the feature interaction relationships among samples within a batch. Specifically, for a given query sample, rather than calculating pairwise similarity, MCR reconstructs it using a linear combination of batch training samples from the other modality to accurately assess cross-modal semantic consistency. Meanwhile, to continuously enhance the feature expression of clean pairs and diminish the impact of noisy ones, we propose a multi-phase iterative reconstruction process. Through this dynamic filtering mechanism, the algorithm progressively eliminates potential noisy samples, therefore effectively achieving the highly robust alignment of features. Furthermore, we introduce a queue mechanism to store high-quality historical samples. This queue provides an expanded dictionary to enrich the global reference space, thereby improving the reliability and accuracy of collaborative representation across batches.

Our main contributions can be summarized as follows:

- We are the first to address noisy correspondence problem through cross-modal collaborative representation, exploring intrinsic semantic associations between modalities.
- We propose **M**ulti-phase **C**ollaborative **R**epresentation (**MCR**), a robust framework for cross-modal retrieval. This framework utilizes reconstruction coefficients to adaptively quantify sample correlations, providing reliable matching confidence for cross-modal alignment.
- Extensive experiments on three widely-used datasets, including Flickr30K, MS-COCO, and Conceptual Captions, demonstrate the effectiveness of our approach.

2 Related Works

2.1 Noisy Correspondence Learning

Noisy correspondence refers to semantically mismatched image-text pairs incorrectly treated as matched samples. NCR [18] first formalized this problem, exploiting the memorization effect of DNNs [9] to identify noisy pairs. Following this, most studies focused on identifying unreliable samples through various strategies. BiCro [42] assessed semantic correlations via bidirectional similarity consistency. GSC [51] distinguished samples by preserving geometric consistency. Alternatively, some works improved stability by designing noise-robust learning objectives. MSCN [10] adopted meta-learning to stabilize similarity estimation under noisy environments. CRCL [33] mitigated noise through complementary learning. CREAM [30] rectified noisy alignments via integrating cross-modal consistency modeling.

Recently, several studies have explored denoising in VLP framework [11, 16, 19]. For CLIP-based NC methods, NPC [50] measured the noise level by evaluating their potential negative impacts. DS-DMR [36] filtered noisy pairs using the CLIP similarity distribution. GLP [22] propagated similarity labels over cross-modal graphs for robust correspondence estimation. However, most methods largely inherit denoising paradigms from noisy label and fail to exploit the rich information within the cross-modal data. Therefore, we propose a framework that deeply mines cross-modal matching relationships for reliable correspondence estimation.

2.2 Collaborative Representation

Collaborative **R**epresentation (**CR**) [48] is a representation learning method designed to capture the latent semantic consistencies and intrinsic relationships of data. Its core idea is to represent a query sample as a linear combination using a dictionary of training samples. This framework, along with extended forms like probabilistic and kernel collaborative representation [1, 15, 28, 40, 52], has been widely applied to visual recognition tasks, including face recognition and image classification [26, 39, 48, 49]. Recently, collaborative representation has also been explored in cross-modal retrieval [47], where test samples are reconstructed using training images and text respectively to yield a unified feature space for cross-modal similarity modeling.

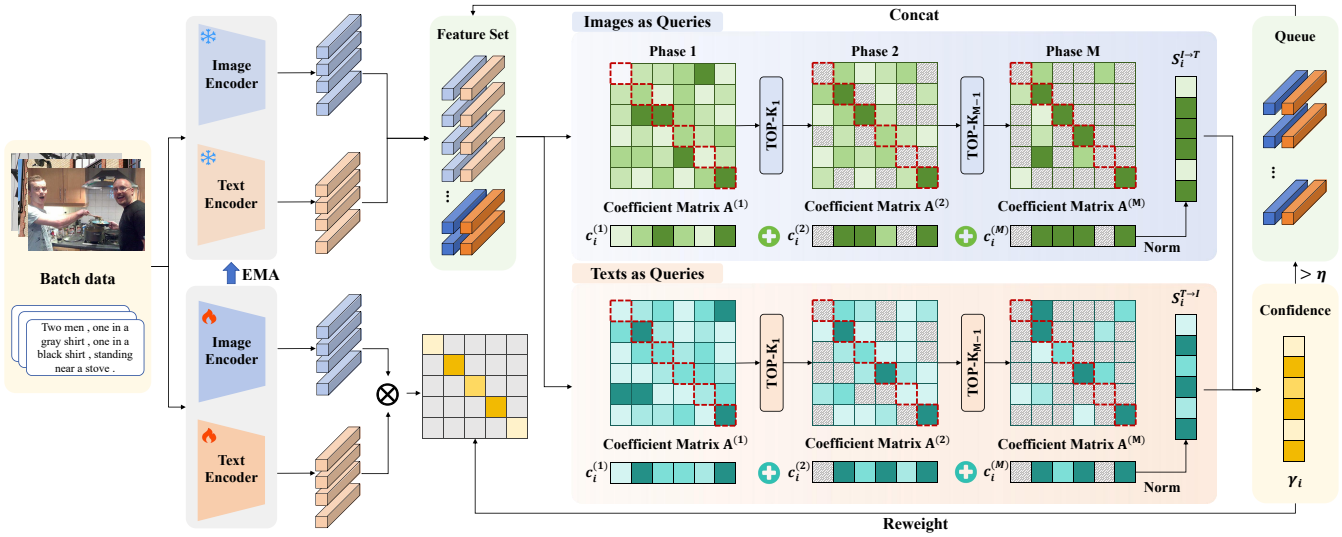


Figure 2: Overview of the proposed MCR. Batch features from EMA encoders are concatenated with historical features from a global queue. In Phase 1, an initial coefficient matrix $A^{(1)}$ is computed to extract diagonal elements $c_i^{(1)}$. Subsequent phases apply Top-K filtering noise to iteratively refine the matrix $A^{(M)}$ and extract $c_i^{(M)}$. Diagonals from all phases are accumulated and normalized to yield bidirectional scores $S_i^{I\to T}$ and $S_i^{T\to I}$, which form the comprehensive confidence weight γ_i for reweighting the loss. High-confidence pairs above η update the global queue.

However, collaborative representation is predominantly applied to traditional unimodal tasks. Its integration into cross-modal retrieval frameworks remains limited, and its potential to resolve NC problems has been unexplored. Therefore, we introduce collaborative representation into the CLIP-based cross-modal retrieval training framework. Specifically, collaborative representation is utilized to estimate the reliability of image-text pairs, thereby progressively enhancing the representations of clean samples while weakening the interference of noise.

3 Methodology

3.1 Problem Definition

Given a cross-modal dataset with noisy correspondences $D = \{(I_i, T_i), y_i\}_{i=1}^N$ containing N training samples. Here, (I_i, T_i) represents the i-th image and text pair, and $y_i \in \{0, 1\}$ indicates the corresponding label ($y_i = 1$ for a match, and $y_i = 0$ for a mismatch).

We adopt the pre-trained vision-language model CLIP [34] as the backbone to build the cross-modal retrieval framework. This model consists of an independent image encoder $f(\cdot)$ and a text encoder $g(\cdot)$. For the matching model parameterized by θ, the primary objective is to project both the visual and textual modalities into a shared representation space. Within this space, the semantic alignment between the modalities is evaluated using the cosine similarity function $Cos(I_i, T_i)$, which is calculated as follows:

$$Cos(I_i, T_i) = \frac{f(I_i) \cdot g(T_i)}{\|f(I_i)\| \cdot \|g(T_i)\|}. \tag{1}$$

During the overall optimization, we employ a symmetric InfoNCE [31] loss function to further promote the semantic alignment of the matched pairs and to increase the distance between the mismatched pairs. This loss function is formally defined as follows:

$$\mathcal{L}_{InfoNCE}(I_i, T_i) = CE(I_i, T_i) + CE(T_i, I_i), \tag{2}$$

$$CE(I_i, T_i) = -\log\left(\frac{\exp(Cos(I_i, T_i)/\tau)}{\sum_{j=1}^{B}\exp(Cos(I_i, T_j)/\tau)}\right), \tag{3}$$

where τ is the temperature parameter that adjusts the scale of the similarity scores, and B is the batch size.

3.2 Collaborative Reconstruction

As illustrated in Fig. 2, given a training batch of size B, we construct the image feature matrix $\boldsymbol{V} = [\boldsymbol{v}_1, \boldsymbol{v}_2, ..., \boldsymbol{v}_B] \in \mathbb{R}^{d\times B}$ and the text feature matrix $\boldsymbol{T} = [\boldsymbol{t}_1, \boldsymbol{t}_2, ..., \boldsymbol{t}_B] \in \mathbb{R}^{d\times B}$ utilizing EMA encoders [13], which are updated via an Exponential Moving Average (EMA) to ensure representation stability. To provide a global reference for coefficient prediction and enhance training stability across batches, we introduce a fixed-capacity queue mechanism. Specifically, we maintain global semantic banks, represented as queue matrices $\boldsymbol{F}_V$ and $\boldsymbol{F}_T$, that store high-quality historical features. During each iteration, we augment the current batch dictionary by concatenating batch features with queue features, forming the expanded dictionaries $\hat{\boldsymbol{V}} = [\boldsymbol{V}, \boldsymbol{F}_V] \in \mathbb{R}^{d\times n}$ and $\hat{\boldsymbol{T}} = [\boldsymbol{T}, \boldsymbol{F}_T] \in \mathbb{R}^{d\times n}$.

The core idea of collaborative representation is to reconstruct a given query sample via a linear combination of all samples within the expanded dictionary from the other modality, thereby evaluating the cross-modal semantic consistency. Taking the Image-to-Text retrieval direction as an example, we utilize the augmented text feature matrix $\hat{\boldsymbol{T}}$ as a dictionary to reconstruct a specific image feature $\boldsymbol{v}_i$. Mathematically, this approximation is expressed as:

$$\boldsymbol{v}_i \approx \hat{\boldsymbol{T}}\boldsymbol{\rho}_i^{(1)}, \tag{4}$$

where $\boldsymbol{\rho}_i^{(1)} = [\rho_{i,1}^{(1)}, \rho_{i,2}^{(1)}, ..., \rho_{i,n}^{(1)}]^\top$ is the collaborative coefficient vector, indicating the reconstruction contribution of each text feature in $\hat{\boldsymbol{T}}$ to the query $\boldsymbol{v}_i$.

To accurately estimate these coefficients, we formulate the approximation as a ridge regression problem by minimizing both the reconstruction residual and an L_2 regularization term. The optimization objective is defined as follows:

$$\min_{\boldsymbol{\rho}_i^{(1)}} \|\boldsymbol{v}_i - \hat{\boldsymbol{T}}\boldsymbol{\rho}_i^{(1)}\|_2^2 + \lambda\|\boldsymbol{\rho}_i^{(1)}\|_2^2, \tag{5}$$

where λ controls the regularization strength. Taking the derivative of the objective function with respect to $\boldsymbol{\rho}_i^{(1)}$ and setting the result to zero yields the following closed-form solution:

$$\boldsymbol{\rho}_i^{(1)} = (\hat{\boldsymbol{T}}^\top\hat{\boldsymbol{T}} + \lambda\boldsymbol{E})^{-1}\hat{\boldsymbol{T}}^\top\boldsymbol{v}_i, \tag{6}$$

where $\boldsymbol{E}$ is an identity matrix with the same dimensions as $\hat{\boldsymbol{T}}^\top\hat{\boldsymbol{T}}$. By formulating the operations for all samples within the batch as a matrix multiplication, we efficiently obtain the complete collaborative representation coefficient matrix:

$$\boldsymbol{A}^{(1)} = (\hat{\boldsymbol{T}}^\top\hat{\boldsymbol{T}} + \lambda\boldsymbol{E})^{-1}\hat{\boldsymbol{T}}^\top\hat{\boldsymbol{V}}. \tag{7}$$

Subsequently, we apply L_1-Normalization along the columns of the matrix $\boldsymbol{A}^{(1)}$ and extract the main diagonal elements of the matrix as the initial confidence scores for the text-reconstructed image at the current phase. This calculation is formulated as follows:

$$c_i^{(1)} = \frac{\boldsymbol{A}_{i,i}^{(1)}}{\sum_{j=1}^{n}|\boldsymbol{A}_{j,i}^{(1)}|}, \tag{8}$$

where $c_i^{(1)}$ quantifies the matching confidence of the i-th pair. A large $c_i^{(1)}$ indicates the text $\boldsymbol{t}_i$ dominates the reconstruction of given image $\boldsymbol{v}_i$, suggesting clean pairs. Conversely, a small $c_i^{(1)}$ implies a marginal contribution, indicating likely noisy ones.

3.3 Multi-phase Iterative Process

While the single-phase collaborative coefficients derived from Eq. (8) provide an initial estimation, they still remain susceptible due to noise information in the large dictionary. Therefore, we selectively filter the dictionary to remove noisy data and reduce the dictionary scale to more accurately estimate the matching confidence of these samples. To accomplish this purpose, we propose a multi-phase iterative elimination mechanism.

Assuming a total of M phases, and m ($2 \le m \le M$) denotes the current phase. For a given image query v_i, the algorithm selects the top K_{m-1} samples via the coefficient vector $\boldsymbol{\rho}_i^{(m-1)}$ obtained from the previous phase. This process constructs a local high-confidence dictionary index set $\Omega_i^{(m)}$ for the query:

$$\Omega_i^{(m)} = \arg\,\mathrm{TopK}(\boldsymbol{\rho}_i^{(m-1)}, K_{m-1}). \tag{9}$$

Guided by this index set, we discard noisy features with marginal contributions and retain a purified text feature subset $\hat{\boldsymbol{T}}_{\Omega_i^{(m)}}$ to serve as the reduced reconstruction dictionary. We then recalculate the collaborative representation coefficients within this local subspace and obtain the updated representation coefficient matrix $\boldsymbol{A}^{(m)}$:

$$\boldsymbol{A}^{(m)} = \left(\hat{\boldsymbol{T}}_{\Omega_i^{(m)}}^\top\hat{\boldsymbol{T}}_{\Omega_i^{(m)}} + \lambda\boldsymbol{E}\right)^{-1}\hat{\boldsymbol{T}}_{\Omega_i^{(m)}}^\top\hat{\boldsymbol{V}}. \tag{10}$$

Algorithm 1 Multi-phase Collaborative Representation Algorithm

Input: Training set $D = \{(I_i, T_i)\}_{i=1}^N$
Initialize main parameters θ, global queues $\boldsymbol{F}_V, \boldsymbol{F}_T$
for each epoch $e = 1, 2, ..., E$ **do**
  for each mini-batch $\{(I_i, T_i)\}_{i=1}^B$ from D **do**
    Extract batch features $\boldsymbol{V}, \boldsymbol{T}$ using EMA encoders
    Concatenate $\boldsymbol{V}, \boldsymbol{T}$ with $\boldsymbol{F}_V, \boldsymbol{F}_T$ to obtain $\hat{\boldsymbol{V}}, \hat{\boldsymbol{T}}$
    // Collaborative Reconstruction
    Calculate initial coefficient matrix $\boldsymbol{A}^{(1)}$ by Eq. (6)- (7)
    Calculate confidence $c_i^{(1)}$ by Eq. (8)
    // Multi-phase Iterative Process
    for phase $m = 2$ to M **do**
      Refine dictionary and calculate $\boldsymbol{A}^{(m)}$ by Eq. (9)- (10)
      Calculate confidence $c_i^{(m)}$ by Eq. (8)
    end for
    // Optimization Objective
    Compute $S_i^{I\to T}$ by Eq. (11)
    Compute $S_i^{T\to I}$ symmetrically
    Calculate matching confidence γ_i by Eq. (12)- (13)
    Update main parameters θ by minimizing Eq. (14)
    Update EMA encoders
    Enqueue highly reliable features by Eq. (15)
  end for
end for
Output: Robust matching model θ

Subsequently, we extract the normalized main diagonal elements as scores $c_i^{(m)}$ at the current phase. We define the truncation value as a monotonically decreasing sequence $\{K_1, K_2, \ldots, K_{M-1}\}$. Through this multi-phase iterative reconstruction, the scores of clean pairs are progressively amplified, whereas noisy samples are gradually excluded from the dictionary to cause score decay.

Finally, we accumulate the scores extracted across all M phases and apply Max-Normalization within the current mini-batch to map them into the $[0, 1]$ range. The resulting cumulative confidence score $S_i^{I\to T}$ for the image-to-text direction is formulated as:

$$S_i^{I\to T} = \frac{\sum_{m=1}^{M} c_i^{(m)}}{\max_{1\le j\le B}\left(\sum_{m=1}^{M} c_j^{(m)}\right)}. \tag{11}$$

Given the symmetry of the modalities, we analogously derive the multi-phase collaborative representation for the text-to-image direction and compute the cumulative confidence score $S_i^{T\to I}$. Since unidirectional evaluation is susceptible to bias, relying solely on one direction may yield suboptimal estimations. To robustly estimate the reliability of samples, we formulate the comprehensive confidence weight γ_i by averaging the bidirectional scores:

$$\gamma_i = (S_i^{I\to T} + S_i^{T\to I})/2. \tag{12}$$

Since γ_i integrates bidirectional cross-modal matching confidence, it can be utilized to distinguish clean and noisy pairs, therefore alleviating the noisy correspondence problem. Specifically, $\gamma_i \to 1$ indicates a semantically aligned clean pair, whereas $\gamma_i \to 0$ signifies a mismatched noisy pair.

Table 1: Image-Text Retrieval performance under different noise ratios of 20%, 40%, and 60% on Flickr30K and MS-COCO 1K.

Noise	Methods	Flickr30K							MS-COCO 1K						
		Image⟶Text			Text⟶Image				Image⟶Text			Text⟶Image			
		R@1	R@5	R@10	R@1	R@5	R@10	rSum	R@1	R@5	R@10	R@1	R@5	R@10	rSum
20%	NCR [18]	75.0	93.9	97.5	58.3	83.0	89.0	496.7	77.7	95.6	98.2	62.6	89.3	95.3	518.7
	DECL [32]	77.5	93.8	97.0	56.1	81.8	88.5	494.7	77.5	95.9	98.4	61.7	89.3	95.4	518.2
	BiCro [42]	78.1	94.4	97.5	60.4	84.4	89.9	504.7	78.8	96.1	98.6	63.7	90.3	95.7	523.2
	PC^2 [6]	78.7	94.9	96.9	59.8	83.9	89.6	503.8	77.8	95.7	98.4	62.8	89.7	95.3	519.7
	L2RM [12]	77.9	95.2	97.8	59.8	83.6	89.5	503.8	80.2	96.3	98.5	64.2	90.1	95.4	524.7
	GSC [51]	78.3	94.6	97.8	60.1	84.5	90.5	505.8	79.5	96.4	98.9	64.4	90.6	95.9	525.7
	ESC [43]	79.0	94.8	97.5	59.1	83.8	89.1	503.3	79.2	**97.0**	**99.1**	64.8	90.7	96.0	526.8
	CLIP [34]	82.3	95.5	98.3	66.0	88.5	93.5	524.1	75.0	93.1	97.2	58.7	86.1	97.2	507.3
	NPC [50]	87.3	97.5	98.8	72.9	92.1	95.8	544.4	79.9	95.9	98.4	66.3	90.8	98.4	529.7
	GLP [22]	88.1	98.2	99.4	74.1	93.3	96.3	549.4	81.0	95.9	98.4	66.4	90.9	98.4	531.0
	MCR	**88.6**	**98.5**	**99.6**	**75.6**	**93.8**	**96.8**	**552.9**	**82.8**	96.8	98.8	**68.7**	**92.0**	**98.8**	**537.9**
40%	NCR [18]	73.5	92.6	95.8	55.7	80.3	86.9	484.8	76.6	95.6	98.2	61.0	88.9	94.9	515.2
	DECL [32]	72.7	92.3	95.4	53.4	79.4	86.4	479.6	75.6	95.5	98.3	59.5	88.3	94.8	512.0
	BiCro [42]	74.6	92.7	96.2	55.5	81.1	87.4	487.5	77.0	95.9	98.3	61.8	89.2	94.9	517.1
	PC^2 [6]	75.8	93.5	96.9	57.5	81.9	88.2	493.8	77.4	95.8	98.4	62.1	89.4	95.1	518.2
	L2RM [12]	75.8	93.2	96.9	56.3	81.0	87.3	490.5	77.5	95.8	98.4	62.0	89.1	94.9	517.7
	GSC [51]	76.5	94.1	97.6	57.5	82.7	88.9	497.3	78.2	95.9	98.2	62.5	89.7	95.4	519.9
	ESC [43]	76.1	93.1	96.4	56.0	80.8	87.2	489.6	78.6	**96.6**	**99.0**	63.2	90.6	95.9	523.9
	CLIP [34]	76.2	93.3	96.5	59.4	85.0	90.9	501.3	70.7	91.7	96.2	54.7	83.4	96.2	492.9
	NPC [50]	85.6	97.5	98.4	71.3	91.3	95.3	539.4	79.4	95.1	98.3	65.0	90.1	98.3	526.2
	GLP [22]	87.9	98.1	99.4	73.8	92.9	96.5	548.6	80.2	95.8	98.5	66.6	90.8	98.5	530.4
	MCR	**88.4**	**98.4**	**99.5**	**74.6**	**93.0**	**96.8**	**550.7**	**81.7**	96.5	98.8	**68.0**	**91.5**	**98.8**	**535.5**
60%	NCR [18]	70.0	91.0	94.4	52.3	76.9	84.0	468.6	72.6	93.8	97.4	57.0	86.4	93.6	500.8
	DECL [32]	65.2	88.4	94.0	46.8	74.0	82.2	450.6	73.0	94.2	97.9	57.0	86.6	93.8	502.5
	BiCro [42]	67.6	90.8	94.4	51.2	77.6	84.7	466.3	73.9	94.4	97.8	58.3	87.2	93.9	505.5
	PC^2 [6]	70.8	90.3	94.4	53.1	79.0	85.9	473.5	74.2	94.4	97.8	58.9	87.5	93.8	506.6
	L2RM [12]	70.0	90.8	95.4	51.3	76.4	83.7	467.6	75.4	94.7	97.9	59.2	87.4	93.8	508.4
	GSC [51]	70.8	91.1	95.9	53.6	79.8	86.8	478.0	75.6	95.1	98.0	60.0	88.3	94.6	511.7
	ESC [43]	72.6	90.9	94.6	53.0	78.6	85.3	475.0	77.2	95.1	98.1	61.1	88.6	94.9	515.0
	CLIP [34]	66.3	87.3	93.0	52.1	78.8	87.4	464.9	67.0	88.8	95.0	49.7	79.6	95.0	475.1
	NPC [50]	83.0	95.9	98.6	68.1	89.6	94.2	529.4	78.2	94.4	97.7	63.1	89.0	97.7	520.1
	GLP [22]	88.2	98.0	99.2	72.2	**92.2**	**96.2**	546.0	80.5	95.9	98.5	66.5	90.6	98.6	530.6
	MCR	**88.7**	**98.4**	**99.5**	**73.3**	**92.2**	**96.2**	**548.3**	**80.6**	**96.3**	**98.8**	**66.6**	**90.9**	**98.8**	**532.0**

3.4 Overall Objective

To mitigate training instability caused by fluctuations across different iterations, we apply the temporal ensembling technique [25] to smoothly update γ_i. Specifically, the final matching confidence $\gamma_i^{(t)}$ at step t is updated via a momentum-based combination:

$$\gamma_i^{(t)} = \beta\gamma_i^{(t-1)} + (1-\beta)\gamma_i^{(t)}, \tag{13}$$

where β is a momentum parameter controlling the smoothing. The smoothed weight γ_i provides a more stable matching confidence of the i-th sample pair.

Finally, we utilize γ_i as a reliable confidence weight to reweight the InfoNCE loss, emphasizing the contribution of clean pairs and suppressing the negative impact of noisy ones. In the experiment, we find that noisy confidence weights reduce to approximately zero, which makes them negligible in the optimization. The overall loss function of MCR is formulated as follows:

$$\mathcal{L}_{Re-InfoNCE} = \frac{1}{B}\sum_{i=1}^{B}\gamma_i\mathcal{L}_{InfoNCE}(I_i, T_i). \tag{14}$$

Following the optimization of the current batch, we leverage the high-confidence weights γ_i to dynamically update the global queues. The purpose of this operation is to continuously select and enqueue highly reliable features from the current batch, thereby providing a broader and purer dictionary space for collaborative reconstruction. The queue is initialized as empty and managed by FIFO strategy, governed by a confidence threshold η:

$$\begin{aligned} \boldsymbol{F}_V^{(t)} &= \{\boldsymbol{v}_i \mid \gamma_i > \eta,\ \boldsymbol{v}_i \in \boldsymbol{V}\}, \\ \boldsymbol{F}_T^{(t)} &= \{\boldsymbol{t}_i \mid \gamma_i > \eta,\ \boldsymbol{t}_i \in \boldsymbol{T}\}. \end{aligned} \tag{15}$$

Algorithm 1 summarizes our proposed MCR.

Table 2: Image-Text Retrieval performance on MS-COCO 5K.

Noise	Methods	Image⟶Text			Text⟶Image			rSum
		R@1	R@5	R@10	R@1	R@5	R@10	
40%	NCR [18]	55.5	82.4	90.2	39.7	68.5	79.2	415.5
	BiCro [42]	56.3	83.0	90.8	40.1	69.0	79.5	418.7
	ESC [43]	56.2	83.2	90.7	41.0	69.5	79.8	420.4
	NPC [50]	61.1	84.8	90.7	44.7	72.1	81.7	435.1
	GLP [22]	64.8	86.7	92.6	47.6	74.4	83.4	449.5
	MCR	**65.2**	**87.0**	**92.7**	**48.2**	**75.0**	**83.8**	**451.9**
60%	NCR [18]	49.6	78.1	87.3	35.5	64.2	75.7	390.4
	BiCro [42]	52.5	80.0	88.4	37.8	66.2	77.1	402.0
	ESC [43]	53.4	81.1	89.2	38.2	66.7	77.5	406.1
	NPC [50]	59.7	82.9	89.7	43.0	70.2	79.9	425.4
	GLP [22]	63.1	**85.9**	**92.0**	46.4	73.7	82.9	444.0
	MCR	**63.2**	85.6	**92.0**	**46.5**	**73.9**	**83.0**	**444.2**

Table 3: Comparison with baselines on CC120K.

Methods	Image⟶Text			Text⟶Image			rSum
	R@1	R@5	R@10	R@1	R@5	R@10	
CLIP [34]	68.8	87.0	92.9	67.8	86.4	90.9	493.8
NPC [50]	71.1	92.0	**96.2**	73.0	90.5	94.8	517.6
GLP [22]	72.0	90.9	94.9	72.2	**90.7**	94.3	515.0
MCR	**72.7**	**92.3**	95.6	**73.9**	**90.7**	**95.1**	**520.3**

4 Experiments

4.1 Datasets and Evaluation Metrics

Datasets. Following previous works [18], the proposed MCR is evaluated on three benchmark datasets: Flickr30K [44], MS-COCO [23], and CC120K [50].

- **Flickr30K** comprises 31,783 images, each image is annotated with 5 captions. Following previous works [18], we utilize 29,783 images for training, 1,000 for validation, and 1,000 for testing.
- **MS-COCO** contains 123,287 images, with 5 annotated captions per image. Following previous works [18], we employ 113,287 images for training, 5,000 for validation, and 5,000 for testing. The evaluation is performed either on the complete test set of 5,000 images or is averaged across 5-fold of 1,000 images.
- **CC120K** is a subset of the web-crawled Conceptual Captions [35], with about 3%-20% mismatched image-text pairs. It consists of 120,851 images, each with a single caption. Following NPC [50], we utilize 118,851 images for training, 1,000 for validation, and 1,000 for testing.

Evaluation Metrics. We employ Recall@K (R@K) to evaluate the cross-modal retrieval performance, which measures the proportion of relevant items found within the top K results of a ranked list. Additionally, to provide a more comprehensive analysis, we report R@1, R@5, and R@10 for both image-to-text and text-to-image retrieval tasks, and summarize these metrics as rSum for a comprehensive performance evaluation.

Table 4: Comparison of methods with ViT-B/32 backbone on noisy MS-COCO.

Noise	Methods	1K R@1	5K R@1	1K rSum
20%	VSE∞ [2]	72.0	51.4	520.2
	PCME++ [3]	70.8	49.5	522.4
	PAU [21]	71.4	51.7	521.5
	NPC [50]	73.1	53.8	529.8
	GLP [22]	73.7	55.2	531.0
	MCR	**75.8**	**57.6**	**537.9**
50%	VSE∞ [2]	38.5	18.4	390.5
	PCME++ [3]	65.7	44.0	503.9
	PAU [21]	69.3	49.6	513.4
	NPC [50]	71.3	51.9	523.4
	GLP [22]	72.9	54.0	528.4
	MCR	**74.4**	**55.8**	**534.3**

Table 5: Effectiveness of each component on Flickr30K with 40% noise. Baseline is CLIP. γ_i^* represents the single-phase weight, β denotes the smoothing technique, Que indicates the queue mechanism, and Enc refers to the EMA encoders.

Components					Image⟶Text			
γ_i	γ_i^*	β	Que	Enc	R@1	R@5	R@10	rSum
Baseline					76.2	93.3	96.5	266.0
✓		✓	✓	✓	**88.4**	**98.4**	99.5	**286.3**
✓			✓	✓	87.6	98.1	99.5	285.2
✓		✓		✓	87.8	**98.4**	99.6	285.8
✓		✓	✓		86.3	97.6	99.5	283.4
	✓	✓	✓	✓	87.4	98.2	**99.7**	285.3
✓					85.1	97.3	98.9	281.3
					Text⟶Image			
Baseline					59.4	85.0	90.9	235.3
✓		✓	✓	✓	**74.6**	**93.0**	**96.8**	**264.4**
✓			✓	✓	73.1	92.3	96.4	261.8
✓		✓		✓	74.6	92.9	96.7	264.2
✓		✓	✓		73.3	92.3	96.0	261.6
	✓	✓	✓	✓	73.1	92.5	96.0	261.6
✓					71.3	91.0	95.1	257.4

4.2 Implementation Details

We adopt the pre-trained CLIP [34] with ViT-B/32 as our backbone and train MCR on a single RTX 3090 GPU. We employ a batch size of 256 and an AdamW [27] optimizer with a weight decay of 0.2. In all experiments, we train the model for 5 epochs with a learning rate of $2e-7$. For the hyperparameters, the momentum for updating the EMA encoders is 0.99. The regularization parameter λ and the confidence threshold η are set to 0.05 and 0.9. The smoothing momentum β and the queue capacity Q_{len} are set to 0.9 and 100.

4.3 Comparison with State-of-the-Arts

We evaluate the proposed MCR against State-of-the-Arts baselines on three datasets. The baselines include SGRAF-based noise-robust methods, such as NCR [18] , DECL [32], BiCro [42], PC^2 [6], L2RM [12] , GSC [51], and ESC [43], and CLIP-based approaches,

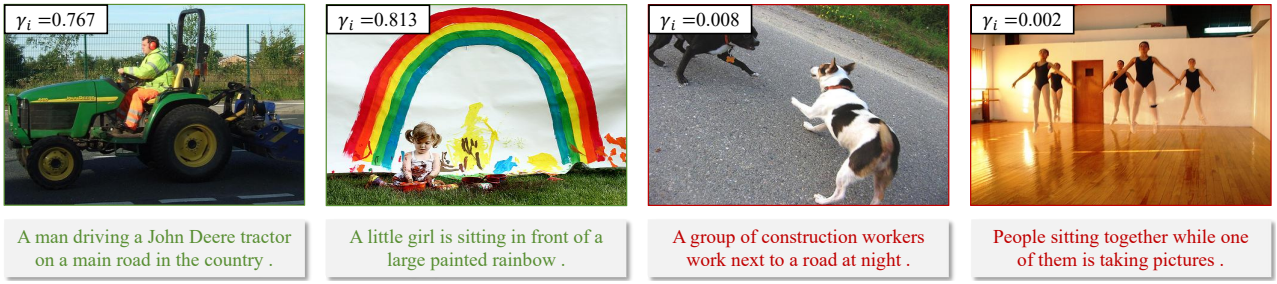


Figure 3: Case study about matching confidence weights on Flickr30K with 40% noise ratio.

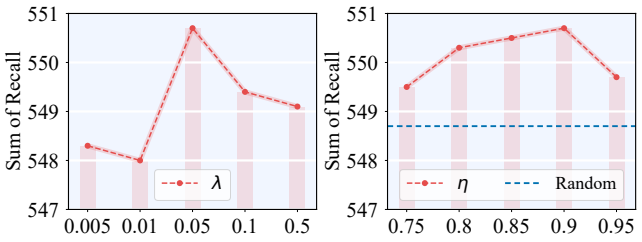


Figure 4: Performance under different hyper-parameters of MCR on Flickr30K with 40% noise. λ is the regularization parameter, and η denotes the enqueue threshold.

including fine-tuned CLIP [34], NPC [50], and GLP [22]. Following the protocol in [18], we simulate noisy correspondence on Flickr30K and MS-COCO by randomly shuffling captions to introduce noise ratios of 20%, 40%, and 60%. Additionally, we validate performance under real-world noise using the CC120K dataset. All reported results are evaluated on the test set using the optimal checkpoint from the validation set.

Results on Flickr30K and MS-COCO. As detailed in Table 1, the proposed method significantly outperforms the other baseline methods across most evaluation metrics. Most importantly, MCR consistently achieves the optimal rSum scores and exhibits high performance under all noise ratios. It is particular that on MS-COCO 1K, MCR's performance under 60% noise (532.0%) even surpasses the performance of GLP under 20% noise (531.0%). Notably, MCR also reaches the highest R@1 scores for both image-to-text and text-to-image retrieval across all noise rates on these two datasets. We also demonstrate the performance of MCR on the full MS-COCO 5K dataset in Table 2. Specifically, MCR outperforms both GLP and NPC. These results demonstrate the superior robustness and competitiveness of MCR against existing approaches.

Results on CC120K. To further validate the comprehensive performance in real-world scenarios, we evaluate the method on the CC120K dataset. As observed in Table 3, MCR achieves the optimal performance, reaching an rSum of 520.3%. Compared to GLP (515.0%) and NPC (517.6%), MCR yields an improvement of 5.3% and 2.7%. Overall, MCR indicates the effectiveness in handling real-world NCs, establishing a strong method for cross-modal retrieval tasks on the CC120K dataset.

Comparison with ViT-B/32 Backbone Methods. Given that the CLIP ViT-B/32 backbone inherently outperforms other architectures, we follow previous works [22, 50] and compare MCR with other methods adopting the same backbone to ensure a fair performance evaluation. We conduct comparative experiments under 20% and 50% noise correspondence scenarios, and report mean R@1 on MS-COCO 1K and 5K, as well as rSum on 1K. As demonstrated in Table 4, MCR still outperforms other methods. Specifically, it surpasses GLP by 6.9% and 5.9% under 20% and 50% noise, respectively. These results demonstrate that even with the same backbone, MCR effectively exploits cross-modal relationships and mitigates noisy correspondence, leading to superior retrieval performance.

Table 6: Experimental results of different truncation sequences K_m on Flickr30K. K_m determines the filtering intensity of the dictionary.

Noise	K_m	Image⟶Text			Text⟶Image			rSum
		R@1	R@5	R@10	R@1	R@5	R@10	
40%	{64, 32, 16, 8}	87.9	98.3	99.4	74.5	93.0	96.7	549.8
	{64, 32, 16, 8, 4}	88.1	98.2	99.5	**74.7**	93.1	**96.8**	550.4
	{128, 64, 32, 16, 8}	88.0	98.3	**99.8**	**74.7**	**93.2**	96.7	**550.7**
	{128, 64, 32, 16, 8, 4}	**88.4**	**98.4**	99.5	74.6	93.0	**96.8**	**550.7**
60%	{64, 32, 16, 8}	87.5	97.9	99.4	72.8	92.3	96.0	545.9
	{64, 32, 16, 8, 4}	87.2	97.3	98.8	73.1	92.1	96.1	544.6
	{128, 64, 32, 16, 8}	87.8	97.7	99.3	73.1	**92.5**	**96.4**	546.8
	{128, 64, 32, 16, 8, 4}	**88.7**	**98.4**	**99.5**	**73.3**	92.2	96.2	**548.3**

4.4 Further Analysis

4.4.1 Ablation Study. We conduct ablation studies on Flickr30K with 40% noise to validate the contributions of key components. From Table 5, we observe that removing any single component degrades retrieval performance, indicating all components are essential for improving robustness.

Specifically, compared to relying solely on the initial single-phase collaborative reconstruction scheme (γ_i^*), the proposed multi-phase strategy (γ_i) significantly improves performance by reinforcing clean samples and progressively weakening noise interference.

Furthermore, removing smoothing technique (β) negatively impacts performance, showing its necessity in preventing training instability caused by fluctuating confidence scores across iterations. Similarly, maintaining a reliable reference space relies on both the queue mechanism (Que) and EMA encoders (Enc), omitting either would compromise the stability or capacity of the dictionary required for robust reconstruction.



Phase 1

Rank 1: [GT] A water bird standing at the ocean 's edge. **0.0178**

Rank 2: One bird opens its mouth while the other bird looks back at it. **0.0149**

Rank 3: Four young men in shorts are walking along a beach on a partly cloudy day. **0.0147**

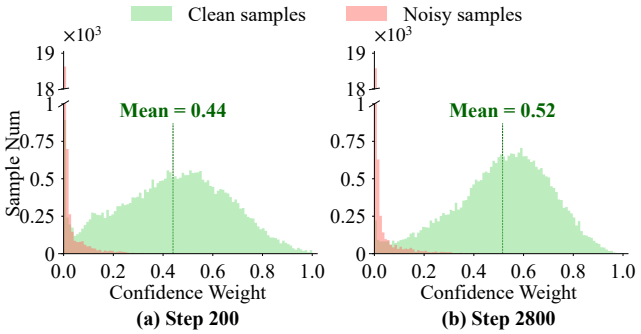
Phase M

Rank 1: [GT] A water bird standing at the ocean 's edge. **0.4688** ⬆

Rank 2: Four young men in shorts are walking along a beach on a partly cloudy day. **0.1765**

Rank 3: A boy wearing white shorts , walks through the water. **0.0873**

Figure 5: Visualize examples about Image-to-Text coefficients evolution on Flickr30K training set under 40% noise.



(a) Step 200 **(b) Step 2800**

Figure 6: Distribution of confidence weights for clean and noisy samples on Flickr30K under 40% noise.



Figure 7: Matching results of Text-to-Image coefficients evolution on Flickr30K under 40% noise.

4.4.2 Hyperparameter Analysis. We evaluate the effects of three key hyperparameters, λ, η and K_m, on retrieval performance. A small λ makes coefficients unstable, while a large one weakens discrimination. A low η enqueues unreliable pairs, whereas a high one slows dictionary updates.

Fig. 4 demonstrates that the model delivers stable performance for $\lambda \in [0.05, 0.1]$ and achieves optimal results at $\lambda = 0.05$. Meanwhile, the retrieval performance reaches its optimal at $\eta = 0.9$. Replacing the confidence-aware queue with a random queue leads to a performance decline, which shows the gains rely on dictionary purity rather than just an expanded dictionary capacity.

We further investigate the impact of K_m on retrieval performance. K_m determines the filtering intensity of the dictionary. As demonstrated in Table 6, varying K_m has a relatively minor impact on the overall performance. The reason is that smoothing technique effectively enhances global training stability, thereby preventing the framework from over-relying on specific sequence designs. Nevertheless, an overly aggressive reduction in K_m disrupts the smoothing process, which can lead to a slight performance degradation under 60% noise. Therefore, we intentionally adopt a progressive decay strategy.

4.5 Visualization Analysis

To further analyze the impact of our method, we visualize various detailed results to provide an intuitive explanation. Fig. 6 illustrates the confidence weight distributions of clean and noisy sample pairs at different training steps on Flickr30K with 40% noise. We can observe that the noisy sample confidence weights tend to settle in zero, while clean samples have remarkably larger weights. Moreover, as training progresses, the weight distributions of clean and noisy samples exhibit a separation trend. These phenomena reveal the noise identification ability of our approach.

Furthermore, we conduct the case study to intuitively demonstrate the capabilities of MCR. Fig. 3 verifies that the model accurately identifies the sample matching status and assigns appropriate confidence weights (γ_i). Specifically, the weights of noisy samples are less than 10^{-2}. This observation is consistent with the histogram in Fig. 6, where the weights of noisy samples are small enough to suppress their effect in the model optimization.

To explore the internal iterative mechanism, Fig. 5 and Fig. 7 visualize the multi-phase coefficient evolution for both Image-to-Text and Text-to-Image reconstructions. In the initial phase, reconstruction weights are less discriminative among the top 3 candidates. However, as the dictionary shrinks across iterations, MCR effectively filters out semantically irrelevant samples, significantly enhancing the weights of original clean sample pairs.

5 Conclusion

In this paper, we address the inevitable noisy correspondence problem in cross-modal retrieval. We propose a novel framework MCR, which introduces a new perspective of cross-modal collaborative representation for robust alignment. Specifically, MCR leverages reconstruction coefficients to estimate the semantic correlations between samples within feature space. Furthermore, a multi-phase iterative process is designed to progressively enhance clean representations and dynamically filter out noisy ones, yielding reliable confidence weights for reweighting the InfoNCE loss. Extensive experiments on three widely-used benchmarks validate the effectiveness and robustness of MCR under both synthetic and real-world noisy correspondence scenarios.

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